

FINAL LAB EXAM

BCDA 5P126

MACHINE LEARNING LAB

MINI-PROJECT CHALLENGE

Think like a company. Build like a data team. Present like a consultant.

Student Project Guide

11 Groups • End-to-End Machine Learning • Role-Based Evaluation

1. Project Overview

Rather than executing isolated code cells in a disconnected exam session, each group operates as an independent AI & Analytics Consultancy firm. Your team has been hired by an enterprise client to tackle a mission-critical business problem from Unit V (Vision, Speech, NLP, Fraud, Diagnostics, Churn, or Forecasting).

The 12-Step Enterprise ML Lifecycle

Every consultancy must deliver an unbroken, reproducible pipeline:

1. Problem Definition & Operational Framing → 2. Data Acquisition & Source Provenance → 3. Data Ingestion & Sanitization → 4. Exploratory Data Analysis & Anomaly Detection → 5. Feature Engineering & Selection → 6. Leakage-Free Validation Strategy → 7. Baseline vs. Advanced ML Modeling → 8. Multi-Metric Evaluation & Calibration → 9. Error Diagnostics & Limitation Analysis → 10. Analytics Engine Integration → 11. Interactive BI Dashboard → 12. Client Pitch & Operational Recommendations.

2. Team Roles

Every team member must formally register under exactly one functional role. Individual grading is tied to your direct code authorship, personal viva voce defense, and your specific role deliverable.

Functional Role	Core Responsibilities	Primary Owned Deliverables	Viva Voce Focus
Data Engineer (DE)	Acquires, validates, and cleans raw data; handles missing/outlier records; builds scalable preprocessing; prevents data leakage.	• Data pipeline script/module • Train/test feature matrices (`.csv`) • Data quality log & audit sheet	Missing value logic, schema validation, data drift handling, leakage prevention.
Data Analyst (DA)	Conducts visual and statistical EDA; detects distributions and anomalies; discovers non-trivial feature correlations.	• Visual EDA section in notebook • Distribution & skewness plots • Top 3 actionable data insights	Correlation interpretation, feature interaction insights, class distribution impact.
Data Scientist (DS)	Formulates modeling strategy; trains baseline vs. candidate ML algorithms; performs hyperparameter tuning & cross-validation.	• Model training & tuning script • Cross-validation comparison table • Serialized model file (`.pkl` / `.onnx`)	Algorithm justification, bias-variance tradeoff, hyperparameter search strategy.
ML Engineer (MLE)	Builds reproducible inference pipeline; structures input-to-prediction code; manages model saving, latency, and operational logging.	• End-to-end inference script (`.predict.py`) • Environment & dependency manifest • Pipeline sanity check tests	Inference latency, pipeline reproducibility, deployment monitoring, containerization readiness.

Analytics Engineer (AE)	Connects cleaned data, model predictions, and business KPIs; formulates decision threshold rules and derived financial metrics.	• Derived metric transformation script • Confusion cost-matrix model • Decision threshold tuning sheet	Connecting probabilities to business thresholds, financial impact of Type I vs Type II errors.
BI / Power BI Developer (BI)	Designs an interactive, executive-ready dashboard (Power BI preferred; Streamlit/Dash accepted) tracking operational KPIs.	• Interactive dashboard (`.pbix` / `app.py`) • Executive KPI summary cards • What-if simulation / single-record tester	Visual hierarchy, executive usability, slicer interactivity, dynamic real-time querying.

Every member must understand the overall project, even though the viva will focus on the assigned role.

3. Group-Wise Topics and Dataset Links

Use the assigned dataset. If a dataset is too large for classroom hardware, use a documented, justified subset.

Grp	Client & Domain	Business Problem	Assigned Dataset	Primary Metrics	Technical Directives & Safeguards	Dataset
G1	SecureVision AI (Physical Security)	Classify incoming surveillance frames into 10 target categories.	CIFAR-10 Dataset	Macro F1-Score, Inference Latency	Use CNN / Transfer Learning. Use a documented subset if GPU/RAM constrained.	CIFAR-10
G2	CallConnect Speech (Customer Support)	Classify customer voice audio into emotional/intent states.	RAVDESS Speech Audio	Classification Accuracy, Confusion Matrix	Extract MFCC / Mel-spectrogram features. Never execute raw uncompressed audio in memory.	RAVDESS Speech
G3	MailGuard Cyber (Enterprise Email)	Filter incoming enterprise messages into Spam vs. Ham.	UCI SMS Spam Collection	Spam Precision, PR-AUC	Prioritize Spam Precision: False Positives (losing important mail) are catastrophic.	UCI SMS Spam Collection
G4	EndpointShield (Cybersecurity)	Classify malicious vs. benign binaries without running them.	EMBER Malware Benchmark	Malware Recall, ROC-AUC	STRICT SAFETY: Use static features only. Never execute live malware files.	EMBER Malware Benchmark
G5	PayShield Fintech (Digital Banking)	Detect fraudulent credit card charges in real-time streams.	Credit Card Fraud Dataset	PR-AUC, F1-Score, Cost per Fraud Miss	Heavily imbalanced (0.17% fraud). Accuracy is unacceptable; evaluate Precision-Recall curve.	Credit Card Fraud Dataset
G6	MediCare Diagnostics (Clinical Health)	Educational risk screening for coronary heart disease.	UCI Heart Disease	Recall (Sensitivity), Log-Loss	EDUCATIONAL USE ONLY. False Negatives (missing sick patient) represent unacceptable clinical risk.	UCI Heart Disease

G7	EchoMedia Analytics (Entertainment)	Classify customer reviews into positive/negative sentiment.	Stanford IMDb Reviews	Balanced Accuracy, Macro F1-Score	Demonstrate text vectorization (TF-IDF / Embeddings) and out-of-vocabulary handling.	Stanford IMDb Large Movie Review Dataset
G8	TelcoPlus Networks (Telecom SaaS)	Predict subscriber churn 30 days prior to contract lapse.	IBM Telco Churn	ROC-AUC, Lift Score, Customer CLV Saved	Connect churn risk to actionable retention offers. Explain model predictions with feature importance.	IBM Telco Customer Churn
G9	CrediTrust Capital (Retail Banking)	Predict default risk (Good vs. Bad credit applicants).	UCI Statlog German Credit	Cost-Weighted F1, Expected Loss	Penalize Type II errors (granting credit to defaulters) 5x higher than Type I errors.	UCI Statlog German Credit Data
G10	Apex Realty AI (Real Estate Tech)	Predict residential housing market prices and deviations.	California Housing	RMSE, MAE, Residual R ²	Continuous regression. Provide residual diagnostics and identify systemic geographic over-predictions.	California Housing
G11	MetroTransit Fleet (Smart Mobility)	Forecast hourly bike-share demand for fleet rebalancing.	UCI Bike Sharing	RMSE, MAPE, Hourly Peak Error	Time-series aware validation. Enforce strict temporal ordering: no future-leakage cross-validation!	UCI Bike Sharing Dataset

4. Mandatory Workflow

- Problem definition and success criteria
- Dataset acquisition and source documentation
- Data understanding: shape, types, target and statistics
- Data cleaning: missing values, duplicates and invalid records
- EDA: distributions, relationships, trends and outliers
- Feature engineering / encoding / selection
- Train-test or validation strategy; prevent data leakage
- Baseline and ML model building
- Model evaluation using appropriate metrics
- Model interpretation and error analysis
- Dashboard / visualization
- Business recommendations and limitations

5. Technical Requirements

- Python is recommended for the main implementation.
- Use suitable libraries such as Pandas, NumPy, Matplotlib, Scikit-learn and/or PyTorch/TensorFlow.
- Explain important preprocessing decisions.
- Compare models where practical.
- Do not rely on accuracy alone for imbalanced classification.
- Document sampling and assumptions.

- Keep the notebook reproducible and organized.
- Discuss limitations and model errors.

Recommended Metrics

- Classification: Accuracy, Precision, Recall, F1-score, Confusion Matrix, ROC-AUC where appropriate.
- Imbalanced classification: Precision, Recall, F1 and PR-AUC/ROC-AUC; discuss false positives/negatives.
- Regression: MAE, MSE, RMSE, R^2 and residual/error analysis.
- Image/audio/text: include class-wise metrics and confusion matrix where feasible.

6. Dashboard Requirement

- Power BI is preferred.
- Include 3–6 useful KPIs.
- Show important patterns/segments.
- Include at least one ML outcome/prediction visual.
- Use meaningful filters/slicers.
- Add a Key Insights section.
- Avoid decorative charts that do not answer a question.

7. Recommended Folder Structure & Submission Requirements

Each team must host their project on a dedicated GitHub repository named **ML-Final-Lab-Group-XX**. The repository link must be submitted on the portal before the exam begins.

Required Submission Directory Tree

```

project_group_XX/
├── README.md           <- Project summary, environment, execution commands
├── requirements.txt     <- Exact Python packages and version locks
├── data/
│   ├── raw/           <- Original pristine dataset source files
│   └── processed/      <- Cleaned, transformed feature matrices
├── notebooks/
│   └── ML_Project.ipynb <- Complete reproducible pipeline (EDA to evaluation)
├── src/
│   ├── preprocessing.py <- Production data transformation functions
│   └── model.py         <- Serialized model training and inference scripts
├── dashboard/
│   └── Project_Dashboard.pbix <- Power BI file (or app.py for Streamlit/Dash)
├── report/
│   └── Project_Report.pdf <- Formal technical report (4–6 pages)
├── presentation/
│   └── Client_Pitch.pptx <- Executive pitch deck (Maximum 7 slides)

```

Mandatory GitHub Protocol & Academic Integrity

- **Large File Handling & GitHub Limits:** Never push files larger than 50MB directly to GitHub. If your dataset exceeds this, store the raw file externally (Google Drive / Kaggle link), add ``data/raw/*`` to your ``.gitignore``, place a 500-row sample in ``data/raw/sample.csv``, and provide a Python script (``download_data.py``) in ``data/raw/`` to fetch the complete data automatically.
- **Individual Commit Accountability:** Every single team member must commit their own code from their own GitHub account. Evaluators will inspect GitHub Commit Insights and PR histories during the viva. Committing all work under one member's account will result in a zero mark for unverified members.
- **README.md as Executive Landing Page:** The root ``README.md`` serves as the public landing page for your consultancy. It must prominently display: (1) Team Name & Member Roster with assigned roles; (2) Client Persona & Problem Statement; (3) Primary Target Metric & Baseline Performance; (4) Step-by-step reproduction commands (``git clone``, ``pip install -r requirements.txt``, and running the scripts).
- **Freeze Tagging (v1.0-final-submission):** At the deadline, create a GitHub Release / Git Tag titled ``v1.0-final-submission``. Any commits pushed after the examination deadline timestamp will not be considered during evaluation.

8. Final Lab: Client Pitch

- 1 min – Business problem and objective
- 2 min – Dataset and EDA findings
- 2 min – ML models and results
- 2 min – Dashboard and recommendations
- 3 min – Q&A / individual viva

Present to the faculty as if they are the client. Do not simply read the notebook.

9. Viva Questions by Role

During the viva, the client will cross-examine students directly. Memorized code will not suffice; you must defend architectural trade-offs:

You may Expect Questions Similar to this but not exactly these questions.

- **Data Engineer:** “Where did the dataset originate? What exact cleaning transformations did you execute and why? How did you ensure test features did not leak into training scaling? How would you automate this pipeline for batch ingestion?”
- **Data Analyst:** “State three non-obvious insights discovered during EDA. Which single feature exhibited the strongest correlation with the target? Which visual chart directly informed the team's feature selection?”
- **Data Scientist:** “Why did you choose this algorithm family over simpler linear baselines? Why is raw accuracy an insufficient metric for your domain? How did you address the bias-variance tradeoff during hyperparameter tuning?”
- **ML Engineer:** “Walk us through your end-to-end inference script. How did you serialize and reload the model artifact? What is your inference latency per transaction/image? What production telemetry would you monitor for data drift?”
- **Analytics Engineer:** “How did you convert raw model probability outputs into concrete operational decisions? What specific decision threshold was chosen, and what are the financial trade-offs of Type I vs. Type II errors?”
- **BI Developer:** “Why were these 4 specific KPIs selected for executive leadership? Walk us through a scenario where an operations manager uses slicers to isolate a critical business anomaly. How do you trigger automated alert views?”

10. Assessment Rubric – 50 Marks

Assessment Category	Marks	Detailed Technical & Operational Expectations
Problem Definition & Workflow	4	Operational framing, clear business objectives, end-to-end pipeline modularity.
Data Acquisition, Cleaning & Pipeline	6	Automated pipeline, missing value handling, outlier mitigation, zero data leakage.
EDA & Analytical Insights	4	Non-trivial statistical charts, correlation analysis, distributions, actionable findings.
ML Modeling Methodology & Tuning	6	Baseline comparison, justifiable algorithm selection, k-fold cross-validation, tuning.
Evaluation, Metrics & Error Analysis	4	Domain-appropriate metric selection, confusion matrix, residual checks, trade-off analysis.
Power BI / Interactive Dashboard	4	3-6 executive KPIs, dynamic slicers, live prediction/inference visual, clean UI.
Pitch Quality & Adherence to Time	2	Crisp slide deck, professional delivery, seamless transitions, strict 10-min adherence.

Part B: Individual Role Viva & Defense (20 Marks per Student)

Individual Evaluation Component	Marks	Specific Assessment Focus
Direct Code Authorship & Artifact Ownership	10	Verifiable proof of personally authoring scripts, notebooks, transforms, or UI views.
Viva Voce Technical Defense & Q&A	10	Ability to defend architectural choices, explain mathematical concepts, and handle client edge cases.

11. Rules and Academic Expectations

- Every member must contribute meaningfully.
- Every member must understand the overall project.
- Do not copy a complete project from another source/group.
- If AI tools are used, students remain responsible for correctness and understanding.
- Cite datasets, references and third-party resources.
- Medical project is educational only, not a clinical diagnostic system.
- For malware, use static features only; never execute malware.
- Do not expose private/confidential data.
- Document assumptions, sampling and limitations.
- Never manipulate test data to improve the score.

12. Suggested Report Structure

- Title Page
- Problem Statement and Business Context
- Dataset Description
- Cleaning and Preprocessing
- EDA
- Feature Engineering
- ML Methodology
- Model Evaluation and Comparison
- Dashboard
- Key Findings and Recommendations
- Limitations and Future Work
- Individual Contributions
- References

13. What Makes a Strong Project?

- Specific problem statement.
- Clear understanding of the dataset.
- Justified preprocessing.
- EDA that produces real insights.
- Reasoned algorithm selection.
- Metrics suited to the problem.
- Error/limitation analysis.
- Dashboard that tells a story.
- Evidence-based recommendations.
- Confident individual viva.

14. Dataset-Specific Notes

- **CIFAR-10:** Image processing may require a manageable subset depending on hardware.
- **RAVDESS:** Audio feature extraction may be time-consuming; document any subset used.
- **SMS Spam:** Focus on text cleaning and vectorization.
- **EMBER:** Static features only; never execute malware.
- **Credit Card Fraud:** Highly imbalanced classification; prioritize precision/recall/F1.
- **Heart Disease:** Educational exercise only; discuss small dataset and limitations.
- **IMDb:** Text preprocessing/vectorization are central.
- **Telco Churn:** Connect predictions to customer-retention actions.
- **German Credit:** Discuss the consequences of classification errors.
- **California Housing:** Use regression metrics and residual analysis.
- **Bike Sharing:** Consider seasonality/time and avoid future leakage.

17. Direct Dataset Links

- **Group 1 – Image Recognition – Object Classification:** [CIFAR-10](#)
- **Group 2 – Speech Recognition – Emotion Classification:** [RAVDESS Speech](#)
- **Group 3 – Email/SMS Spam Filtering:** [UCI SMS Spam Collection](#)
- **Group 4 – Malware Filtering:** [EMBER Malware Benchmark](#)
- **Group 5 – Online Fraud Detection:** [Credit Card Fraud Dataset](#)
- **Group 6 – Medical Diagnosis – Heart Disease:** [UCI Heart Disease](#)
- **Group 7 – Sentiment Analysis:** [Stanford IMDB Large Movie Review Dataset](#)
- **Group 8 – Customer Churn Prediction:** [IBM Telco Customer Churn](#)
- **Group 9 – Credit Risk Prediction:** [UCI Statlog German Credit Data](#)
- **Group 10 – House Price Prediction:** [California Housing](#)
- **Group 11 – Demand Prediction – Bike Sharing:** [UCI Bike Sharing Dataset](#)